Chapter 1F: Heart Failure

Medvale Internal Medicine - Cardiovascular Medicine

Original educational chapter. Uploaded cardiology source material was used as a coverage checklist, especially for HFrEF/HFpEF framing, NYHA class, diagnostic tests, CXR/BNP/echo, and treatment anchors. Current guideline concepts are integrated where helpful.

Why this matters clinically

Heart failure is not a single disease. It is a clinical syndrome in which the heart cannot deliver adequate forward blood flow, cannot accept venous return without high filling pressures, or both. The patient may present with dyspnea, edema, fatigue, pulmonary congestion, renal dysfunction, arrhythmias, or sudden decompensation after a seemingly small trigger. For a learner, the most important first association is that heart failure converts a cardiac problem into a whole-body problem: high venous pressure congests the lungs, liver, gut, kidneys, and legs, while low forward output activates neurohormonal systems that initially help perfusion but eventually worsen remodeling and fluid retention.

The uploaded cardiology section frames congestive heart failure as the final common pathway for many cardiac diseases and highlights that both systolic and diastolic dysfunction often coexist. It also emphasizes that HFrEF is associated with reduced ejection fraction, that HFpEF is associated with impaired filling or ventricular stiffness, and that the clinical exam should look for left-sided congestion, right-sided congestion, and evidence of poor perfusion.

In practice, heart failure questions usually ask the clinician to do one of five things: identify HF as the cause of dyspnea, classify the phenotype by ejection fraction, find the precipitating trigger, treat congestion safely, or choose mortality-reducing therapy. This chapter is therefore written as an association-rich map: symptom to physiology, test to interpretation, ejection fraction to treatment, and timing to decision.

Core definitions and ejection fraction categories

Ejection fraction is the percentage of left ventricular end-diastolic volume ejected with each beat. It does not directly measure total cardiac output, and a patient can have severe symptoms with a preserved EF if filling pressures are high. Still, EF is clinically central because it guides prognosis, therapy, device decisions, and exam classification.

Older teaching materials often divide systolic dysfunction as EF below 40% and preserved systolic function as EF above 40%. Modern guideline terminology is more specific: HFrEF is EF 40% or lower, HFmrEF is EF 41-49%, HFpEF is EF 50% or higher, and HFimpEF describes a patient whose EF was previously 40% or lower and later improves above 40%. The important clinical point is that HFrEF has the clearest mortality-reducing drug pathway, while HFpEF management focuses heavily on congestion relief, blood pressure control, atrial fibrillation management, ischemia treatment, obesity/sleep apnea/comorbidity management, and selected disease-modifying therapy.

Category	EF anchor	Core idea	Treatment association
HFrEF	<=40%	Impaired systolic contraction; reduced pump function	Four-pillar GDMT: ARNI/ACEi/ARB, evidence-based beta blocker, MRA, SGLT2 inhibitor
HFmrEF	41-49%	Borderline/mildly reduced systolic function; often behaves closer to HFrEF as EF declines	SGLT2 inhibitor has strong role; consider other HFrEF-type therapies when appropriate

Category	EF anchor	Core idea	Treatment association
HFpEF	>=50%	Impaired relaxation, stiffness, high filling pressures despite preserved EF	Diuretics for congestion; SGLT2 inhibitor; manage BP, AF, ischemia, obesity, CKD, sleep apnea
HFimpEF	Previously <=40%, now >40%	Recovered/improved EF does not mean cured myocardium	Continue HFrEF therapy to reduce relapse risk

Numbers to know: heart failure

Heart failure has many numbers, but only a subset tends to drive bedside reasoning. EF cutoffs classify phenotype. BNP and NT-proBNP help separate cardiac from noncardiac dyspnea. Daily weight changes suggest early fluid accumulation. Sodium and fluid targets help structure lifestyle therapy. EF 35% is a recurring device and mineralocorticoid antagonist anchor in classic teaching and many guideline pathways.

Topic	Number anchor	Why it matters
HFrEF	EF <=40%	Defines reduced systolic function in current guideline terminology
Older source systolic dysfunction	EF <40%	Uploaded source uses reduced EF below 40% as the systolic dysfunction anchor
HFmrEF	EF 41-49%	Mildly reduced range; treatment overlaps with HFrEF, especially SGLT2 inhibitors
HFpEF	EF >=50%	Preserved EF; filling pressures and stiffness drive symptoms
MRA/device anchor	EF <=35%	Classic high-yield threshold for spironolactone/eplerenone in selected patients and ICD consideration
BNP decompensated HF clue	>100 pg/mL	Uploaded source notes BNP above 100 pg/mL correlates with decompensated CHF
Fluid restriction classic target	1.5-2.0 L/day	Used in symptomatic or volume-overloaded HF patients
Sodium restriction classic target	<4 g/day	Uploaded source lists less than 4 g/day; stricter targets may be individualized
Ivabradine heart-rate anchor	>70 bpm	For selected HFrEF patients in sinus rhythm despite maximally tolerated beta blocker
NYHA I	Symptoms only with vigorous activity	Nearly asymptomatic functional class
NYHA II	Symptomswith prolonged/moderate exertion	Slight limitation
NYHA III	Symptoms with ordinary activity	Marked limitation
NYHA IV	Symptoms at rest	Severe/incapacitating HF

Pathophysiology: why compensation becomes decompensation

Heart failure begins with impaired cardiac performance, but symptoms are amplified by compensation. When effective arterial blood flow falls, the sympathetic nervous system and renin-angiotensin-aldosterone system activate. Sympathetic tone raises heart rate and contractility, while RAAS increases vasoconstriction, sodium retention, and water retention. In the short term, this can preserve blood pressure and perfusion. In the long term, it increases afterload, preload, wall stress, myocardial oxygen demand, and ventricular remodeling.

The Frank-Starling relationship explains why extra preload can initially help but later harm. In a normal heart, increasing preload stretches myocardial fibers and increases stroke volume. In a failing heart, the curve is flatter: large increases in venous return produce only small increases in stroke volume, while pulmonary and systemic venous pressures rise. This is why a heart failure patient can gain fluid without meaningfully improving forward flow. The extra fluid becomes edema, orthopnea, crackles, hepatic congestion, ascites, or pleural effusions.

Original diagram: compensation loop in heart failure

Myocardial injury, pressure overload, volume overload, or impaired filling

-> lower effective forward flow or higher filling pressure

-> sympathetic activation + RAAS activation

-> vasoconstriction + sodium/water retention

-> higher afterload + higher preload + remodeling

-> dyspnea, edema, renal dysfunction, arrhythmia risk, decompensation

HFrEF: reduced pump function

Heart failure with reduced ejection fraction is primarily a systolic pump problem. The ventricle cannot contract effectively, so end-systolic volume rises and EF falls. Common causes include ischemic heart disease, prior myocardial infarction, chronic hypertension, dilated cardiomyopathy, myocarditis, alcohol toxicity, methamphetamine or cocaine exposure, anthracycline chemotherapy, radiation injury, valvular regurgitation, Chagas disease, HIV, tachycardia-mediated cardiomyopathy, peripartum cardiomyopathy, and inherited cardiomyopathy. The uploaded source lists ischemic heart disease or recent MI, idiopathic disease, hypertension, myocarditis, drugs, infiltrative disease, radiation, thyroid disease, peripartum cardiomyopathy, infection, valvular disease, high-output states, and congenital/hereditary causes as important etiologies.

The graph-RAG association should be direct: HFrEF -> impaired contractility -> reduced EF -> low forward output -> neurohormonal activation -> fluid retention and remodeling. Because the neurohormonal response drives disease progression, HFrEF treatment is not just diuresis. Diuretics improve congestion and symptoms, but mortality reduction comes from therapies that interrupt maladaptive neurohormonal and renal-glucose signaling pathways: ARNI or ACE inhibitor/ARB, evidence-based beta blocker, mineralocorticoid receptor antagonist, and SGLT2 inhibitor.

HFpEF: preserved EF but high filling pressure

Heart failure with preserved ejection fraction is not 'normal heart function.' It means the fraction ejected is preserved, but the ventricle may be stiff, hypertrophied, slow to relax, or unable to fill without high pressure. The patient can become severely dyspneic because pressure backs up into the left atrium and pulmonary veins even if EF is 50% or higher. HFpEF is strongly associated with aging, chronic hypertension, obesity, diabetes, chronic kidney disease, atrial fibrillation, sleep apnea, and ischemia. The uploaded source specifically emphasizes hypertension leading to myocardial hypertrophy as the most common cause of diastolic dysfunction, and also lists aortic stenosis, mitral stenosis, aortic regurgitation, and restrictive cardiomyopathy causes such as amyloidosis, sarcoidosis, and hemochromatosis.

In HFpEF, small changes in volume, heart rate, rhythm, and blood pressure can produce large symptom changes. Atrial fibrillation is especially important because a stiff ventricle depends on atrial contraction for late diastolic filling. Tachycardia shortens diastole, uncontrolled hypertension raises afterload and filling pressures, and renal dysfunction makes volume control difficult. Treatment therefore centers on relieving congestion with diuretics, controlling blood pressure, treating ischemia, managing atrial fibrillation, addressing obesity and sleep apnea, and using SGLT2 inhibitors when appropriate.

High-output heart failure

High-output heart failure is a different association. The heart may have normal or even increased cardiac output, but the metabolic demand or low systemic vascular resistance is so extreme that the output is still inadequate for the body. The uploaded source lists chronic anemia, pregnancy, hyperthyroidism, arteriovenous fistulas, wet beriberi from thiamine deficiency, Paget disease of bone, MR, and aortic insufficiency among high-output associations. These conditions rarely cause heart failure alone in a normal heart, but they can unmask or worsen HF in a patient with underlying cardiac disease.

For graph-RAG: high-output HF connects to anemia, thyrotoxicosis, AV fistula, pregnancy, beriberi, and low SVR physiology; HFrEF connects to reduced contractility and remodeling; HFpEF connects to stiffness, hypertrophy, impaired relaxation, and high filling pressures.

Clinical features: left-sided heart failure

Left-sided heart failure produces pulmonary venous congestion and low forward output. Dyspnea is the central symptom because elevated left-sided filling pressures transmit backward into the pulmonary circulation, causing interstitial edema and impaired gas exchange. Orthopnea occurs when lying flat increases venous return and redistributes fluid from the legs and splanchnic circulation into the thorax. Paroxysmal nocturnal dyspnea is classically described as sudden awakening after 1-2 hours of sleep with severe shortness of breath, often relieved by sitting or standing. A nocturnal cough can occur for the same reason: pulmonary congestion worsens in the recumbent position.

Poor forward output can also cause confusion, fatigue, diaphoresis, cool extremities, and cyanosis. On exam, left-sided HF may show a displaced point of maximal impulse from cardiomegaly, an S3 gallop from rapid filling into a dilated noncompliant ventricle, an S4 gallop from atrial contraction into a stiff ventricle, crackles from pulmonary edema, dullness to percussion from pleural effusion, and a loud P2 if pulmonary hypertension has developed. The source emphasizes that S3 is among the most specific signs of CHF and that crackles initially begin at the lung bases.

Finding	Association	Mechanism
Dyspnea	Left-sided HF	Pulmonary congestion and interstitial edema
Orthopnea	Advanced congestion	Recumbency increases venous return and thoracic fluid
PND	More specific congestion clue	Awakening 1-2 hours after sleep due to pulmonary edema
S3	Volume overload, dilated ventricle, HFrEF	Rapid filling into noncompliant LV
S4	Stiff ventricle, HFpEF, LVH	Atrial contraction into noncompliant LV
Crackles	Pulmonary edema	Fluid in alveolar/interstitial spaces

Finding	Association	Mechanism
Loud P2	Pulmonary hypertension	Elevated pulmonary arterial pressure

Clinical features: right-sided heart failure

Right-sided heart failure produces systemic venous congestion. It may result from primary right ventricular disease, pulmonary hypertension, chronic lung disease, pulmonary embolic disease, valvular disease, or most commonly from longstanding left-sided heart failure. When left-sided filling pressures remain elevated, pulmonary venous hypertension can lead to pulmonary arterial hypertension, increasing right ventricular afterload and eventually causing right-sided failure.

The classic findings are peripheral pitting edema, jugular venous distention, hepatomegaly or hepatojugular reflux, ascites, and nocturia. Nocturia occurs because lying down at night mobilizes dependent edema back into the circulation, improving renal perfusion and increasing urine production. The source notes that peripheral pitting edema is nonspecific when isolated and is more likely due to secondary venous insufficiency, while JVD, hepatomegaly/hepatojugular reflux, ascites, and right ventricular heave strengthen the right-sided HF association.

NYHA functional classification

NYHA class is a functional language for symptom burden. It does not directly define EF, anatomy, or cause. A patient with HFpEF can be NYHA IV, and a patient with HFrEF can be NYHA I. The classification matters because it communicates daily limitation, tracks progression, and appears in treatment/device discussions.

NYHA class	Functional description	Typical wording
I	No limitation of ordinary activity	Symptoms only with vigorous activity, such as playing sports
II	Slight limitation	Symptoms with prolonged or moderate exertion, such as climbing stairs or carrying heavy packages
III	Marked limitation	Symptoms with ordinary daily activity, such as walking across a room or getting dressed
IV	Symptoms at rest	Incapacitating symptoms or discomfort even without exertion

Diagnostic approach to suspected heart failure

A new patient with suspected heart failure needs confirmation of the syndrome, identification of the phenotype, and evaluation for cause or trigger. The uploaded source suggests CBC, BMP, cardiac enzymes when indicated, BNP, chest x-ray, ECG, and echocardiogram in a new CHF patient. Those tests map well to clinical reasoning: CBC can reveal anemia or infection; BMP identifies renal dysfunction and potassium abnormalities; troponin can detect ischemic injury; BNP supports cardiac congestion; CXR shows cardiomegaly and pulmonary edema; ECG detects ischemia, prior infarct, arrhythmia, chamber enlargement, or conduction delay; and echocardiography defines EF, chamber size, valves, pericardial disease, and pulmonary pressures.

Echocardiography is the key structural test. It separates HFrEF from HFpEF, estimates EF, identifies chamber dilation or hypertrophy, evaluates valvular stenosis or regurgitation, and can reveal regional wall motion abnormalities suggesting ischemic cardiomyopathy. BNP is released from ventricular myocardium in response to increased wall stretch. A BNP above 100 pg/mL is a classic decompensated HF clue, but BNP is not perfect: it may rise with age, renal disease, pulmonary hypertension, and atrial fibrillation, and it may be lower than expected in obesity. The source specifically emphasizes BNP as useful for differentiating dyspnea due to CHF from COPD.

Test	What it tells you	High-yield association
CBC	Anemia, infection	Anemia can precipitate or mimic HF dyspnea
BMP	Creatinine, BUN, potassium, sodium	Renal dysfunction and electrolytes guide diuretics/RAAS therapy
BNP/NT-proBNP	Ventricular stretch	BNP >100 pg/mL supports decompensated HF in classic teaching
Chest x-ray	Congestion, cardiomegaly, alternative lung disease	Kerley B lines, interstitial edema, pleural effusion
ECG	Rhythm, ischemia, infarct, conduction	Often nonspecific but rarely useless
Echocardiogram	EF, valves, chambers, pericardium	Most important initial structural test
Stress test/cath	Ischemic cause	Consider when CAD may be underlying cause
MUGA	Quantitative EF	Useful when precise EF is needed and echo is limited

Chest x-ray patterns

Chest x-ray can support heart failure but should not be used alone to exclude it. Cardiomegaly suggests chronic chamber enlargement. Kerley B lines are short horizontal lines near the lung bases, caused by interlobular septal thickening from interstitial edema. Prominent interstitial markings, perihilar edema, pleural effusions, and pulmonary vascular redistribution all support congestion. However, a patient with flash pulmonary edema can have acute symptoms before dramatic radiographic changes, and a patient with chronic lung disease may have overlapping findings.

For graph-RAG: CXR -> cardiomegaly + Kerley B lines + pleural effusion + pulmonary edema -> supports CHF; echo -> EF and valve diagnosis; BNP -> ventricular wall stretch; BMP -> renal/electrolyte safety during diuresis.

Common precipitants of acute decompensated heart failure

A stable heart failure patient decompensates when a trigger increases demand, decreases contractility, increases volume, raises afterload, causes arrhythmia, or reduces medication effectiveness. The common inpatient question is not simply, 'Does this patient have HF?' but 'Why did they worsen today?' Missing the trigger can lead to repeated admissions.

Precipitant	Mechanism	Clinical clue
Dietary sodium or medication nonadherence	Volume overload	Weight gain, edema, high JVP
Acute coronary syndrome	New ischemic pump dysfunction	Chest pain, troponin rise, ischemic ECG
Atrial fibrillation with RVR	Loss of atrial kick, tachycardia, poor filling	Palpitations, irregular rhythm
Hypertensive emergency	Afterload surge	Flash pulmonary edema
Renal failure	Impaired sodium/water excretion	Rising creatinine, poor diuretic response
Infection	Demand increase, cytokine stress	Fever, leukocytosis, pneumonia/UTI
Anemia	High-output demand, low oxygen delivery	Fatigue, pallor, low hemoglobin

Precipitant	Mechanism	Clinical clue
NSAIDs	Sodium retention, renal vasoconstriction	Worsening edema after new pain med
Alcohol/cocaine/methamphetamine	Toxicity, ischemia, arrhythmia	Substance exposure
Thyroid disease	Metabolic demand or bradycardia effects	Hyperthyroid high-output state or hypothyroid cardiomyopathy

Treatment goals: separate congestion relief from survival therapy

A major heart failure error is to treat the number on the BNP or the EF category while ignoring the immediate physiology. The first treatment question is whether the patient is wet or dry and warm or cold. 'Wet' means congested: pulmonary edema, JVD, edema, ascites, pleural effusion, orthopnea, or elevated filling pressures. 'Cold' means poor perfusion: hypotension, cool extremities, altered mental status, renal failure, narrow pulse pressure, or shock. A wet-warm patient usually needs diuresis and vasodilator strategy if blood pressure allows. A wet-cold patient may need ICU-level care, inotropes, vasopressors, or mechanical circulatory support depending on severity. A dry-cold patient raises concern for overdiuresis, low output, or alternative shock physiology.

Loop diuretics such as furosemide relieve congestion by promoting sodium and water excretion. They improve dyspnea and edema but have not been shown to reduce mortality by themselves. Thiazide-type diuretics such as metolazone can augment loop diuretics in resistant congestion, but this increases risk for hypokalemia, hyponatremia, renal dysfunction, and volume depletion. Therefore the patient needs monitoring of weight, urine output, renal function, potassium, magnesium, and symptoms.

Lifestyle therapy creates daily feedback loops. Sodium restriction reduces fluid retention. The uploaded source lists sodium restriction less than 4 g/day and fluid restriction 1.5-2.0 L/day. Daily weight monitoring helps detect early fluid accumulation before severe dyspnea develops. Smoking cessation, alcohol restriction, exercise programming, weight loss, influenza vaccination, and pneumococcal vaccination reduce triggers and improve resilience.

HFrEF guideline-directed medical therapy

HFrEF is the phenotype with the strongest evidence for mortality-reducing medications. Contemporary GDMT is often described as four pillars: renin-angiotensin system/neprilysin pathway therapy, evidence-based beta blocker, mineralocorticoid receptor antagonist, and SGLT2 inhibitor. The uploaded source predates some current emphasis but still includes ACE inhibitors or ARBs, evidence-based beta blockers, aldosterone antagonists when EF is below 35%, hydralazine/nitrates in selected patients, sacubitril/valsartan, ivabradine when heart rate remains above 70 despite maximally tolerated beta-blocker therapy, and digoxin for symptom reduction without mortality benefit.

Therapy	Mechanism association	High-yield caution
ARNI / ACE inhibitor / ARB	Reduces RAAS-mediated vasoconstriction/remodeling; ARNI also increases natriuretic peptide signaling	Monitor BP, creatinine, potassium; avoid ACEi in angioedema history; washout needed when switching ACEi to ARNI
Evidence-based beta blocker	Blocks sympathetic toxicity, lowers arrhythmia risk, improves remodeling	Use carvedilol, metoprolol succinate, or bisoprolol; avoid starting during shock or severe decompensation
MRA	Blocks aldosterone-mediated sodium retention and fibrosis	Useful in selected EF <=35% patients; monitor potassium and renal function

Therapy	Mechanism association	High-yield caution
SGLT2 inhibitor	Natriuretic/metabolic/renal effects; reduces HF hospitalization and mortality in HFrEF	Monitor volume status, genital infections, ketoacidosis risk in selected patients
Hydralazine + nitrate	Afterload reduction + preload reduction	Especially useful in self-identified Black patients with persistent symptoms on GDMT and when RAAS agents are contraindicated
Ivabradine	Slows sinus node without beta-blocker effect	Selected HFrEF patients in sinus rhythm with HR >70 on maximally tolerated beta blocker
Digoxin	Increases inotropy and slows AV node	Can reduce symptoms/hospitalizations but not mortality; toxicity risk with renal dysfunction/hypokalemia

HFpEF treatment

HFpEF treatment is less about forcing the EF higher and more about lowering filling pressures, controlling comorbidities, and preventing hospitalization. Diuretics are used for congestion, but overdiuresis can reduce preload too much and worsen renal function or dizziness. Blood pressure control is central because chronic hypertension causes LV hypertrophy and stiffness. Atrial fibrillation should be managed because tachycardia and loss of atrial contraction can sharply worsen filling. Ischemia should be treated because stiff myocardium has little reserve. Obesity, diabetes, chronic kidney disease, and sleep apnea should be addressed because they sustain inflammation, volume stress, and pulmonary pressures.

Modern guidance supports SGLT2 inhibitors for many patients with HFpEF to reduce HF hospitalization. MRAs or ARNI may be considered in selected patients, especially when EF is closer to the mildly reduced range, but the evidence is less robust than in HFrEF. The uploaded source correctly emphasizes the older high-yield teaching that HFpEF is treated symptomatically with diuretics, salt/fluid restriction, and daily weights, and that no broad drug class had historically shown a large mortality benefit in randomized controlled trials.

Devices and advanced heart failure

Some HFrEF patients remain at high risk despite optimized medical therapy. An implantable cardioverter-defibrillator is considered for selected patients with persistently low EF, often 35% or lower, because sudden death from ventricular arrhythmia is a major HFrEF mechanism. Cardiac resynchronization therapy is considered when electrical dyssynchrony, especially left bundle branch block with a wide QRS, contributes to inefficient contraction. Advanced HF therapies include referral to a HF specialty team, left ventricular assist device evaluation, transplant evaluation, palliative care integration, and goals-of-care planning.

The uploaded source states a key exam association: the most common cause of death from HFrEF is sudden death from ventricular arrhythmias, and ischemia can provoke ventricular arrhythmias. This links ischemic cardiomyopathy, low EF, ventricular tachyarrhythmias, ICD therapy, and sudden cardiac death prevention.

Original algorithm: approach to dyspnea with suspected heart failure

Heart failure diagnostic and management flow

Dyspnea, edema, orthopnea, PND, fatigue, or pulmonary edema

-> Check vitals, oxygenation, perfusion, JVP, lung exam, edema, ECG

-> If unstable: treat acute pulmonary edema/shock, consider ACS/arrhythmia/PE

-> Send CBC, BMP, BNP/NT-proBNP, troponin when indicated; obtain CXR

-> Echocardiogram to define EF, valves, chambers, pericardium, pulmonary pressures

-> Classify: HFrEF, HFmrEF, HFpEF, HFimpEF; identify precipitant

-> Treat congestion with diuretics if wet; start/optimize phenotype-specific therapy

-> Plan monitoring: weights, symptoms, renal function, potassium, follow-up

Common pitfalls

One pitfall is equating preserved EF with absence of heart failure. HFpEF patients may have severe pulmonary congestion because their stiff ventricles require high filling pressures. Another pitfall is relying on BNP without context. BNP rises with stretch, but it can be affected by renal dysfunction, age, obesity, pulmonary hypertension, and atrial fibrillation. A third pitfall is giving fluids to every patient with rising creatinine. In decompensated HF, renal dysfunction may reflect venous congestion and low forward flow; the patient may need decongestion rather than more fluid.

A fourth pitfall is treating HFrEF only with loop diuretics. Diuretics relieve congestion but do not replace mortality-reducing GDMT. A fifth pitfall is starting or increasing beta-blockers during cardiogenic shock or severe acute decompensation; beta-blockers are lifesaving chronically but can worsen unstable low-output states if introduced at the wrong moment. A sixth pitfall is missing ischemia as a trigger. New HF, worsening HF, or sudden pulmonary edema should prompt consideration of ACS or ischemic cardiomyopathy.

Practice questions

Question 1

A 68-year-old man has progressive dyspnea, orthopnea, bibasilar crackles, an S3 gallop, and leg edema. BNP is 620 pg/mL and CXR shows cardiomegaly with Kerley B lines. What is the most likely syndrome?

Answer and explanation: Decompensated heart failure. BNP elevation, pulmonary congestion, S3, edema, and cardiomegaly support HF. Kerley B lines represent interstitial edema from elevated pulmonary venous pressure.

Question 2

A patient has EF 30% after a prior MI. Symptoms improve with furosemide, but he is not on other therapy. What medication concept is missing?

Answer and explanation: Guideline-directed mortality-reducing HFrEF therapy. Loop diuretics improve congestion but do not replace ARNI/ACEi/ARB, evidence-based beta blocker, MRA, and SGLT2 inhibitor when appropriate.

Question 3

A 74-year-old woman with long-standing hypertension has dyspnea and pulmonary edema. Echo shows EF 60%, concentric LV hypertrophy, and high filling pressures. What type of HF is this?

Answer and explanation: HFpEF. Preserved EF does not exclude HF; hypertension can cause LV hypertrophy and stiff diastolic filling.

Question 4

A patient with HFrEF has worsening edema after starting ibuprofen for knee pain. What is the mechanism?

Answer and explanation: NSAIDs can worsen HF by causing sodium retention and renal vasoconstriction, reducing diuretic effectiveness and increasing congestion.

Question 5

A patient with EF 25% remains symptomatic despite optimized medications. What sudden-death mechanism should you remember?

Answer and explanation: Ventricular arrhythmia. Low-EF HFrEF is associated with sudden death from VT/VF, which is why ICD eligibility matters in selected patients.

Chapter summary

Heart failure is a syndrome of inadequate forward output, elevated filling pressure, or both. HFrEF is reduced systolic function, currently defined as EF 40% or lower, while HFpEF is preserved EF with impaired filling and high filling pressures. HF symptoms connect directly to physiology: left-sided HF causes dyspnea, orthopnea, PND, crackles, S3, pulmonary edema, and pleural effusions; right-sided HF causes JVD, edema, hepatomegaly, hepatojugular reflux, ascites, and nocturia. BNP above 100 pg/mL is a classic decompensated HF clue, but echo is the key structural test because it defines EF, valves, chambers, and pericardial disease. Treatment begins with determining whether the patient is wet or dry and warm or cold. Diuretics relieve congestion, but HFrEF mortality reduction requires GDMT. HFpEF management focuses on decongestion, blood pressure control, rhythm and ischemia management, comorbidity treatment, and selected therapies such as SGLT2 inhibitors. The graph-RAG core is: cardiac disease -> impaired contraction or filling -> neurohormonal activation -> volume retention/remodeling -> congestion and low-output symptoms -> phenotype-specific treatment.

Source notes and references

1. Uploaded cardiology3.pdf was used as a coverage checklist for chest pain differential, CHF general characteristics, HFrEF/HFpEF framing, Frank-Starling physiology, NYHA classification, diagnostic tests, CXR/BNP/echo findings, lifestyle measures, diuretics, HFrEF medications, HFpEF symptomatic management, and sudden ventricular arrhythmia risk in HFrEF.

2. Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure. Circulation. 2022;145:e895-e1032.

3. This chapter is an original educational synthesis for Medvale. It is not a substitute for local protocols, specialist consultation, or current prescribing guidance.